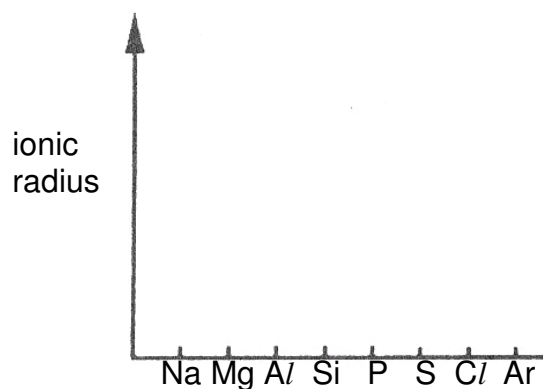
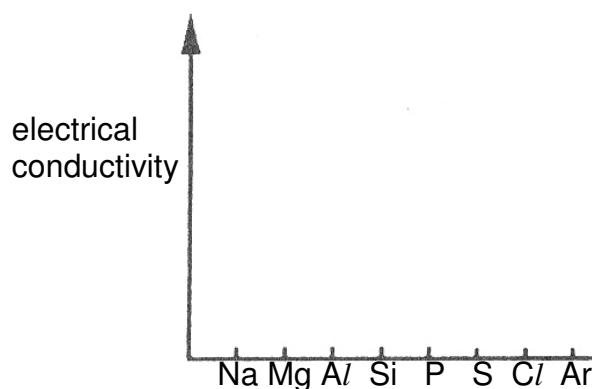
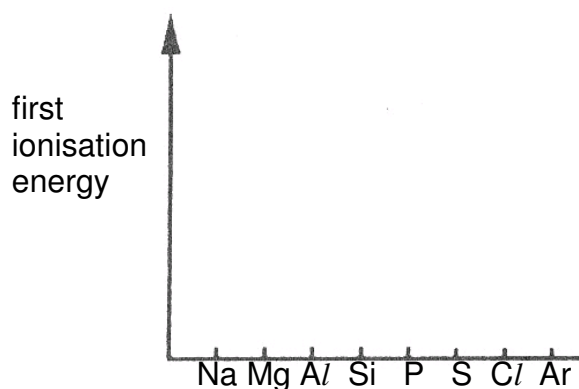


Section A

Answer **all** questions in this section.

1 *Use of the Data Booklet is relevant to this question.*

- (a) (i) Complete these sketches for elements of the third period to show how each property changes along the period.



- (ii) Briefly explain why each property varies as shown by your answers in (i).

first ionisation energy

electrical conductivity

ionic radius

[6]

- (b)** Sodium sulfide (Na_2S) is primarily used in the pulp and paper industry in the Kraft process.

Complete the electronic configuration for the following species.

(i) the Na atom $1s^2$ _____

(ii) the S^{2-} ion $1s^2$ _____

[2]

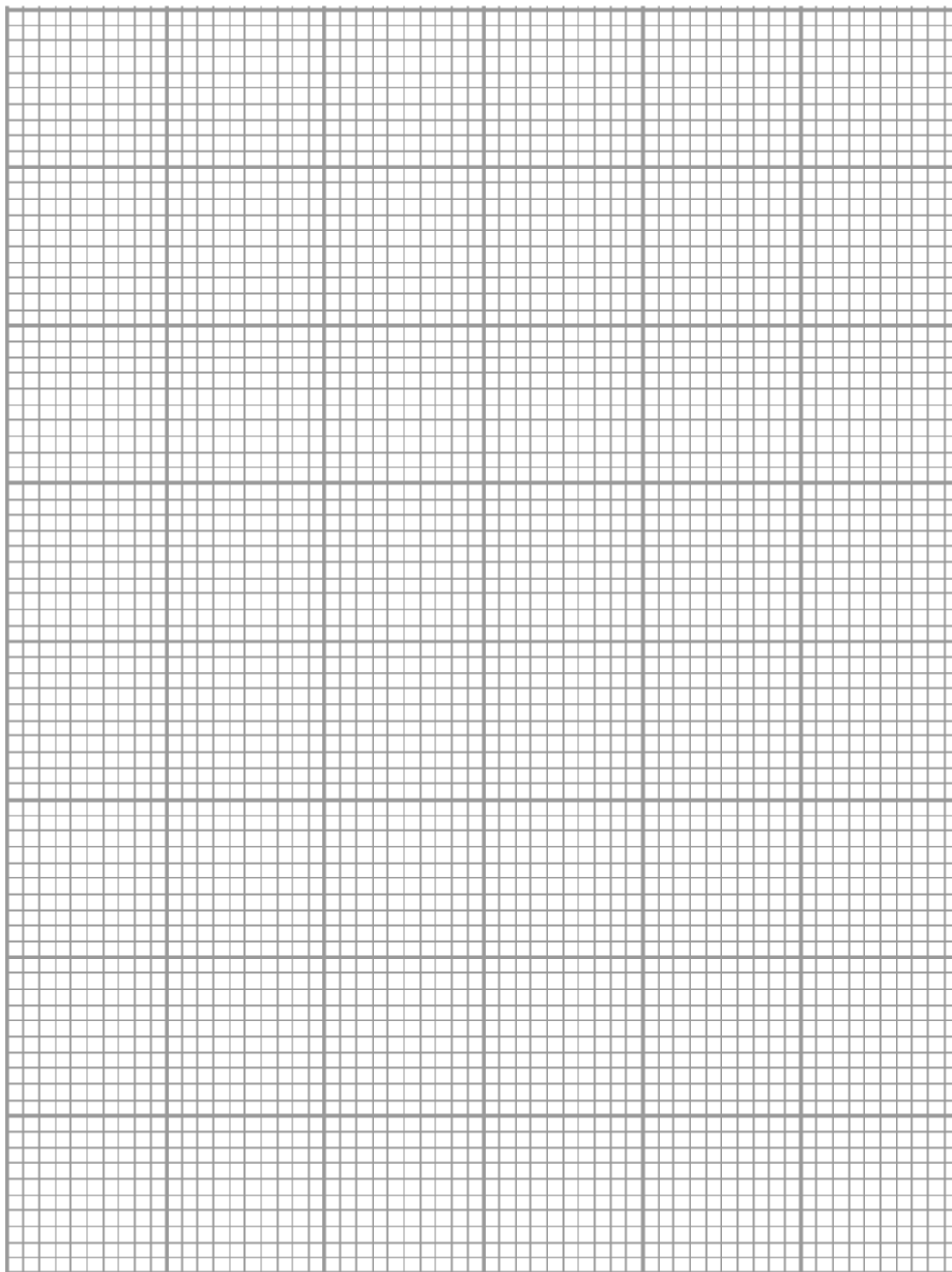
[Total: 8]

2 This question is about the combustion of organic compounds.

- (a) The standard enthalpy changes of combustion of five straight chain alkanes are given in the table below.

alkane	CH ₄	C ₂ H ₆	C ₃ H ₈	C ₄ H ₁₀	C ₅ H ₁₂
$\Delta H_c^\ominus / \text{kJ mol}^{-1}$	-890	-1560	-2230	-2920	-3570

On the grid below, plot a graph to show how the standard enthalpy changes of combustion vary for these five alkanes.



[1]

(b) Use the graph you have plotted in (a), determine

(i) the **average** standard enthalpy change of combustion of a 'CH₂' group;

.....

(ii) the standard enthalpy change of combustion of hydrogen gas;

.....

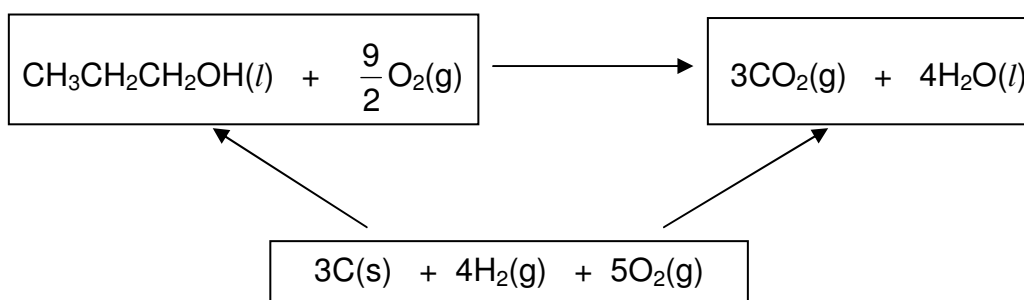
(iii) the standard enthalpy change of combustion of hexane, C₆H₁₄.

.....

[3]

(c) Alcohols are also commonly used as fuels.

The diagram below shows an energy cycle involving propan-1-ol, CH₃CH₂CH₂OH.



Use the following data to calculate the standard enthalpy change of combustion of propan-1-ol.

$$\Delta H_f^\ominus(\text{CO}_2) = -393 \text{ kJ mol}^{-1}$$

$$\Delta H_f^\ominus(\text{H}_2\text{O}) = -286 \text{ kJ mol}^{-1}$$

$$\Delta H_f^\ominus(\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}) = -303 \text{ kJ mol}^{-1}$$

[3]

- (d) (i)** Use the bond energies from the *Data Booklet* to calculate another value for the standard enthalpy change of combustion of propan-1-ol.

- (ii)** Comment on any difference in your answers to **(c)** and **(d)(i)**.

.....

.....

[3]

[Total: 10]

- 3 A student obtained the following results when analysing the compound **A**.

test		observation
test 1	relative molecular mass	72.0
test 2	% composition by mass	C, 66.7%; H, 11.1%; O, 22.2%
test 3	reactions with Br ₂ (aq)	Br ₂ decolourised
test 4	reaction with Na(s)	H ₂ (g) evolved
test 5	reaction with warm Cr ₂ O ₇ ²⁻ /H ⁺	green colour observed

The student allowed test 5 to go to completion and then investigated the **product** of the completed reaction. He obtained the following result:

test 6	reaction with 2,4-dinitrophenylhydrazine	no reaction
--------	--	-------------

- (a) Calculate the molecular formula of **A**.

[2]

- (b) What can be deduced about the nature of **A** in the following tests?

(i) test 3

(ii) test 4

[2]

- (c) (i) What functional group would have given a positive result in test 6?

.....

- (ii) What functional group is shown to be present in **A** by tests 5 and 6?

.....

[2]

- (d) Compound **A** shows *cis-trans* isomerism.

How does *cis-trans* isomerism arise in organic compounds?

.....

.....

..... [2]

- (e) Use all the information above to draw labelled, displayed formulae of all the stereoisomers of compound **A**.

[2]

[Total: 10]

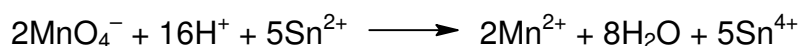
- 4 Sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$, is a common mineral acid used in many chemical reactions. This question is about two main uses of the acid.

(a) Dilute sulfuric acid can be used to dissolve solids for analysis.

Tin forms an oxide, **B**, that contains the metal in both oxidation states II and IV. The formula of **B** can be found by the following method.

- A sample of **B** was dissolved in $\text{H}_2\text{SO}_4(\text{aq})$, producing solution **C**, which contained a mixture of tin(II) sulfate and tin(IV) sulfate.
- A 25.0 cm^3 sample of solution **C** was titrated with $0.0200 \text{ mol dm}^{-3} \text{ KMnO}_4$. 13.50 cm^3 of KMnO_4 was required to reach the end-point.
- Another 25.0 cm^3 sample of solution **C** was stirred with an excess of powdered zinc. This converted all the tin into tin(II). The excess of zinc powder was filtered off and the filtrate was titrated with $0.0200 \text{ mol dm}^{-3} \text{ KMnO}_4$, as before. This time 20.30 cm^3 of KMnO_4 was required to reach the end-point.

The equation for the reaction occurring during the titration is as follows:



- (i) Write a balanced equation for the reaction between Zn and Sn^{4+} .

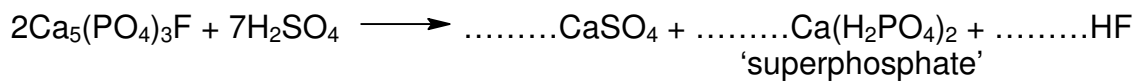
- (ii) Use the results of the two titrations to calculate

- the number of moles of Sn^{2+} in the first titration sample,
- the number of moles of Sn^{2+} in the second titration sample.

- (iii) Use the results of your calculation in (ii) to deduce the $\text{Sn}^{2+}/\text{Sn}^{4+}$ ratio in the oxide **B**, and hence suggest the formula of **B**.

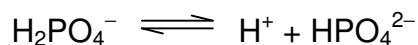
[6]

- (b) One major industrial use of dilute sulfuric acid is to convert phosphate rock (calcium fluorophosphate(V)) into 'superphosphate' for use as a fertiliser. The process can be represented by the following partially balanced equation.



- (i) Balance the equation.
- (ii) Use your balanced equation to calculate the mass of H_2SO_4 required to manufacture 1.0 kg of superphosphate fertiliser.

- (iii) Solutions of hydrogenphosphates are useful buffers for biochemical experiments.



Write two equations to show how the $\text{H}_2\text{PO}_4^-/\text{HPO}_4^{2-}$ system can act as a buffer.

.....

.....

[6]

[Total: 12]

Section B

Answer **two** of the three questions in this section on separate answer paper.

- 5 Despite recent health worries linking aluminum to Alzheimer's disease, aluminum is still used extensively in the modern world.

(a) Aluminium oxide and aluminium chloride are two common compounds of aluminium.

- (i) The melting point of aluminium chloride is 192.4°C while that for aluminium oxide is 2072°C .

Explain the difference in their melting points in terms of the structure and bonding present in each compound.

- (ii) Write balanced equations to show how aluminium oxide reacts with

- dilute hydrochloric acid,
- aqueous sodium hydroxide.

- (iii) At low temperatures, the vapour of aluminium chloride has an empirical formula of AlCl_3 with M_r of 267.

Suggest the molecular formula of the vapour, and draw a suitable diagram to illustrate its bonding.

[6]

- (b) The yellow-white solid of aluminium chloride reacts with water in two separate ways.

- When a few drops of water are added to the solid, steamy white fumes are evolved and a white solid remains, which is insoluble in water.
- When a large amount of water is added to the solid, a clear, weakly acidic solution results.

Write equations, including state symbols, for these two reactions to account for the observations.

[4]

Magnesium, phosphorus and sulfur are elements from the same period as aluminium.

- (c) Describe what would be observed when magnesium and sulfur are **separately** burnt in excess air.

[2]

- (d) The following table shows the lattice energies of magnesium oxide and magnesium chloride.

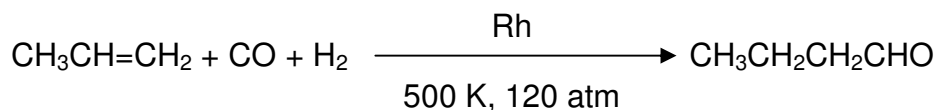
substance	lattice energy / kJ mol^{-1}
magnesium oxide	-3800
magnesium chloride	-2526

- (i) With reference to magnesium oxide, explain the term *lattice energy*.
- (ii) Account for the difference in the numerical values of the lattice energies of the two compounds.
- [3]
- (e) Describe, with an equation, the reaction of phosphorus(V) chloride with water. State the observation when a few drops of Universal Indicator is added to the resulting solution.
- [2]
- (f) Both oxygen and sulfur are from Group VI and can form compounds with fluorine. Oxygen only forms the compound OF_2 , whereas sulfur can form the compounds SF_2 , SF_4 and SF_6 .
- (i) Draw a 'dot and cross' diagram to illustrate the bonding in SF_4 .
You only need to draw the outer electrons.
- (ii) Based on your answer in (i), suggest the shape of the SF_4 molecule.
- (iii) Suggest an explanation for the difference in the number of compounds each element can form with fluorine.

[3]

[Total: 20]

- 6 The OXO reaction is an industrially important process to produce aldehydes from alkenes using carbon monoxide and hydrogen under suitable conditions. The reaction with propene is shown below.



A set of experiments is carried out to investigate the kinetics of the oxo reaction with propene. The following results are obtained.

Experiment	$[\text{CH}_3\text{CH}=\text{CH}_2]$ / mol dm^{-3}	$[\text{CO}]$ / mol dm^{-3}	$[\text{H}_2]$ / mol dm^{-3}	Initial rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	0.250	0.200	0.400	1.67×10^{-3}
2	0.500	0.200	0.400	3.33×10^{-3}
3	0.500	0.300	0.400	5.00×10^{-3}
4	0.500	0.300	0.200	2.50×10^{-3}

- (a) (i) Use the experimental data above to determine the order of reaction with respect to each reactant.
- (ii) Write a rate equation for the reaction.
- (iii) Calculate a value for the rate constant and state its units.

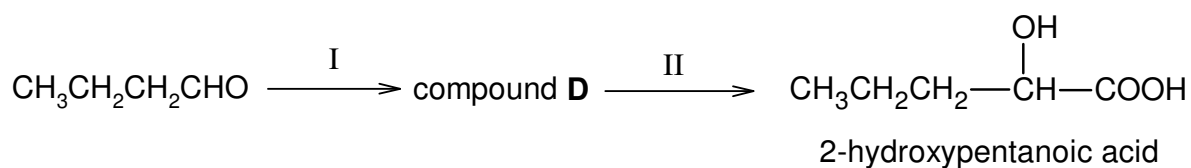
[6]

- (b) Rhodium metal is used as a *catalyst* in the OXO reaction.

- (i) Define *catalyst*.
- (ii) With reference to the Maxwell-Boltzmann energy distribution curve, explain the effect of using rhodium metal in the OXO reaction.

[4]

- (c) 2-hydroxypentanoic acid can be synthesized from butanal in two steps.



- (i) Give the structural formula of compound **D**.
- (ii) State the reagents and conditions required for **each** step, I and II.
- (iii) Write a balanced equation for the reaction that occurs when 2-hydroxypentanoic acid is treated with LiAlH_4 in dry ether.
Your equation should clearly show the structural formulae of **all** organic compounds.
- (iv) When step II is carried out for a prolonged period of time, a compound **E** with the molecular formula $\text{C}_{10}\text{H}_{16}\text{O}_4$ may be produced.
Suggest a possible structure for compound **E**.

[6]

- (d) Describe how you can carry out **two** separate chemical tests to distinguish butanal, butanone, and butanoic acid from one another.

For **each** test, state the reagents and conditions required, and give the observations that would be made with **each** compound.

[4]

[Total: 20]

- 7 (a) Carboxylic acids are important organic compounds which are widely used as food flavourings. Carboxylic acids are usually *weak Brønsted acids*.

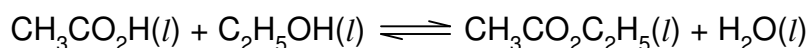
- (i) Explain what is meant by a *weak Brønsted acid*.
- (ii) The K_a values for some organic acids are listed below.

acid	$K_a / \text{mol dm}^{-3}$
$\text{CH}_3\text{CO}_2\text{H}$	1.7×10^{-5}
$\text{ClCH}_2\text{CO}_2\text{H}$	1.3×10^{-3}
$\text{Cl}_2\text{CHCO}_2\text{H}$	5.0×10^{-2}

Explain the trend in K_a values in terms of the structures of these acids.

[5]

- (b) Esters such as ethyl ethanoate can be formed by reacting carboxylic acids with alcohols.



This reaction is an example of a dynamic equilibrium.

- (i) Explain what is meant by the term *dynamic equilibrium*.
- (ii) Write the expression for the equilibrium constant for this reaction, K_c .
- (iii) For this equilibrium, the value of K_c is 4.0 at 298 K.

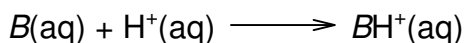
A mixture containing 0.5 mol ethanoic acid, 0.5 mol ethanol, 0.1 mol ethyl ethanoate and 0.1 mol water was set up and allowed to come to equilibrium at 298 K.

Calculate the amount, in moles, of each substance present at equilibrium.

- (iv) State and explain what would happen to the position of the equilibrium when 0.05 mol ethyl ethanoate is withdrawn from the equilibrium mixture in (iii).

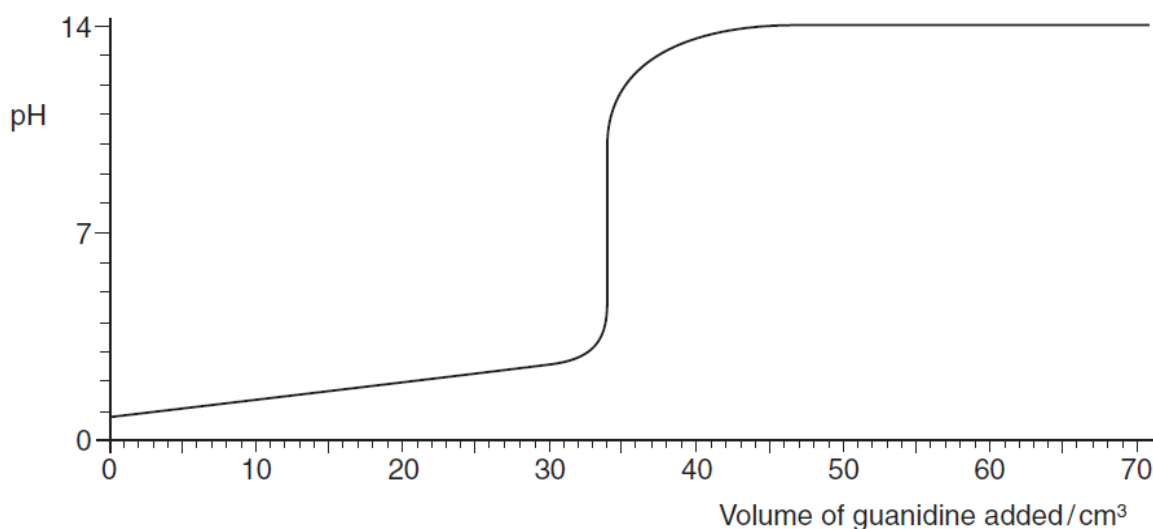
[8]

- (c) Guanidine is an organic base, containing the elements carbon, hydrogen and nitrogen. Its reaction with acids can be represented as follows.



where B represents the molecule of guanidine.

When a 25.0 cm^3 sample of dilute sulfuric acid was titrated against a solution of guanidine, the following titration curve was obtained.



Use this curve to answer the following questions.

- (i) Is guanidine a strong or weak base? Explain your answer.
- (ii) The pH of sulfuric acid at the start of the experiment was 0.70. Calculate $[\text{H}^+]$, and hence the concentration of sulfuric acid, at the start of the experiment.
- (iii) Calculate the concentration of guanidine in mol dm^{-3} .
- (iv) From the list of indicators given below, suggest a suitable indicator that can be used to accurately determine the endpoint of the titration. Give a reason for your choice.

indicator	colour of indicator at		range of pH over which colour change occurs
	pH 1	pH 14	
methyl orange	red	orange	3.1 – 4.4
bromocresol purple	yellow	purple	5.2 – 6.8
trinitrobenzoic acid	colourless	orange-red	12.1 – 13.4

[7]

[Total: 20]